High-fidelity libraries for enzyme optimization

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Abstract

Science abstract target: ∼125 words. Written last, on purpose — it should compress the actual results rather than lead them. The structure below follows the three-move spine: frontier, fragments, PETase. Numbers in `[brackets]` are the ones that must be filled from measurement, not estimated.

- **Hook.** De novo design now produces enzyme backbones faster than they can be optimized. Optimization needs libraries that are large, diverse *and* target-specific; existing methods deliver at most two of the three.
- **Move 1 — the frontier.** GRPO fine-tuning of ProteinMPNN against a structure-prediction reward moves the diversity/fidelity trade-off rather than sliding along it: on a public benchmark, [**3.98**]× the pass rate at matched sequence diversity.
- **Move 2 — the fragments.** Splitting designs at fixed boundaries and recombining across them turns $\sum_i |F_i|$ synthesized parts into $\prod_i |F_i|$ testable variants, and the recombinants retain [**XX**] % of the parental pass rate — a physically buildable library of [**10^X**] members.
- **Move 3 — the demonstration.** Three rounds of closed-loop optimization on a de novo PETase: a pilot library on a fluorogenic probe, a larger library on a PET-proximal substrate, and a third library designed from the accumulated sequence–activity data.
- **Headline.** The best construct is [**250**]× faster than the starting design and exceeds every characterized natural enzyme on the same probe.
- **Implication.** Function-aligned search plus manufacturing-aware library construction plus an experimental closed loop is a general recipe for de novo enzyme optimization.

[TODO: abstract — write once the fragment and PETase numbers land]

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Document index

Every source file in `paper/`, what it holds, and how far along it is. **drafted** means real prose exists; **scaffold** means planning notes only; **needs data** means the section cannot be written until data lands.

File	Contents	Status
Build		
`main.tex`	Compile order; \input's everything	**drafted**
`preamble.tex`	Packages, palette, `\plan` / `\todo` macros, draft-mode switch	**drafted**
`Makefile`	`make pdf`; SVG→PDF via Inkscape	**drafted**
`refs.bib`	Bibliography; all entries machine-retrieved	**drafted**
Front matter		
`abstract.tex`	~125-word abstract, written last	**scaffold**
`intro.tex`	Four-paragraph introduction arc	**scaffold**
`related_work.tex`	Status quo: backbone generation, filtering, RL for inverse folding, library design, the gap	**drafted**
Results — the three-move spine		
`results/frontier.tex`	Move 1: RL moves the diversity/fidelity frontier. Public benchmark, referee-reproducible	**scaffold**
`results/fragments.tex`	Move 2: split + recombine → combinatorial library. **No data yet**	**needs data**
`results/petase.tex`	Move 3: three closed-loop rounds to 250×; the demonstration	**needs data**
Methods		
`methods/overview.tex`	GRPO, fragment libraries, surrogate	**scaffold**
`methods/petase.tex`	Three libraries, surrogate, assays	**scaffold**
Back matter		
`conclusion.tex`	Four-paragraph discussion and outlook	**scaffold**
Not in this build		
`notebook.tex`	Separate document: the E1–E21 record	**drafted**
`experiments.tex`	Arms, selection rules, slate E1–E21, algorithm audit; **\input by notebook.tex, not by main.tex**	**drafted**
`supplementary/*`	Heme and protease results + methods, cut from the spine and parked	**needs data**

The paper and the record are two documents. `main.tex` builds the paper (`make pdf`); `notebook.tex` builds the experimental record (`make notebook`); `make all` builds both. The split is deliberate: the notebook keeps pre-registrations next to the results that falsified them, which is what makes a retraction checkable and what a paper cannot carry and stay readable. A claim moves from the notebook to the paper by being rewritten as prose, never by \input.

Two companion files live in `paper/` but are deliberately *not* part of either build:

- `outline_computational.md` — working brainstorm for a computational-first version of the paper: landscape analysis, the three candidate framings, seven proposed experiments (E1–E7), figure plan, and risk register. Section 2 is the LaTeX rendering of its landscape half.
- `figures/palette.py` — the plotting counterpart to the `cleo*` colors defined in `preamble.tex`.

Draft mode. This document currently compiles with `\draftmodetrue` in `preamble.tex`, which renders all planning notes in gray and all `\todo` markers in red. Setting `\draftmodefalse` hides every one of them, leaving only submission prose — useful for checking how much real text actually exists.

1 Introduction

Science research articles do not use `\section` labels in the main narrative. The headings here exist so this draft has a usable table of contents; strip them at submission. Paragraph scaffolding is kept visible so collaborators can see the intended arc. Target: four paragraphs, ~800 words.

P1 — The promise, and where it stalls

- RFdiffusion and its successors design enzyme backbones from scratch for chemistries with no natural template. Backbone generation is no longer the bottleneck.
- First-pass designs are catalytically weak — typically orders of magnitude below a useful enzyme. The bottleneck moved downstream, to optimization.
- **The framing this paper adopts:** optimization is limited by what you can physically *build and test*, not by what you can compute. That reframes the problem as library construction.

[TODO: p1 — de novo design succeeds at backbones, stalls at activity]

P2 — Three properties, and why nothing delivers all three

- A library that can rescue a weak de novo enzyme has to be simultaneously **large** (the active region is far away), **diverse** (it must cover directions, not just distance) and **target-specific** (every member must still fold and hold the active site). Name these three explicitly here — they are the spine of the whole paper and each results section closes one of them.
- Site-saturation mutagenesis is specific and buildable but strictly local; it cannot reach a variant 30 mutations away.
- Vanilla MPNN re-sampling is structurally informed but functionally blind. Low temperature returns one solution sampled repeatedly; high temperature returns diversity that does not fold. Section 4 measures exactly this trade-off.
- Scale alone is available — **(author?)** [1] reach $\sim 10^{16}$ designs by making the generative model *manufacturing-aware*, factorizing it to match what combinatorial DNA assembly can physically sample. **This is the closest prior art to Move 2 and must be cited as such, not as a foil.** The distinction to draw: they constrain the model to the chemistry from the outset and demonstrate on antibody repertoires, where a natural distribution supplies the objective; we train for catalytic geometry first and factorize afterwards, because for a de novo enzyme there is no natural distribution to imitate and the reward has to be built.
- Cross-reference Section 2 for the full treatment.

[TODO: p2 — the three properties; SSM, MPNN resampling, and the scale-only approaches. Cite weinstein2026manufacturing here as nearest prior art]

P3 — What we contribute

- **(i) A frontier shift, not a knob.** GRPO fine-tuning of ProteinMPNN against a structure-prediction reward, with a KL anchor to the base model that holds the output distribution open. Measured on a public benchmark against the full temperature ladder.
- **(ii) Combinatorial realization.** Fragment splitting and recombination convert the trained policy's output into a library that is exponentially larger than the number of DNA parts synthesized, while keeping the pass rate the policy bought.
- **(iii) A closed loop that actually closed.** Three rounds on a de novo PETase, each library designed using the measured activity of the last.
- Frame (i) and (ii) as *method* and (iii) as *demonstration*. Resist the temptation to present three co-equal applications — the earlier draft did, and it diluted all three.

[TODO: p3 — three contributions]

P4 — Headline numbers + roadmap

- Frontier: [**3.98**]× pass rate at matched diversity on the AME benchmark; U_{15} [**808**] vs [**608**] for the best baseline temperature.
- Fragments: [**N**] parts → [**10^X**] reachable variants at [**XX**] % retained pass rate.
- PETase: [**250**]× over the starting design; best variant ∼[**30**] mutations out; exceeds characterized natural enzymes on the same probe.
- One-sentence roadmap of the three results sections.

[TODO: p4 — headline results and roadmap]

2 Status quo: where the field stands and where the gap is

This section is written as real prose (not scaffold) because it is the part of the paper that does not depend on our own unfinished data. Every citation here was verified against Crossref, the arXiv API, or Europe PMC on 2026-08-04; the retrieval procedure is documented at the top of `refs.bib`. Quantitative claims attributed to other groups are taken verbatim from their abstracts.

2.1 Backbone generation is solved well enough that sequence design is now the bottleneck

Deep generative models now produce enzyme backbones for a wide range of chemistries. RFdiffusion established de novo design of protein structure and function by denoising diffusion over backbone coordinates [2]. RFdiffusion2 removed the requirement that catalytic geometry be specified at the residue level: it designs enzymes directly from functional-group geometries without specifying residue order or performing inverse rotamer generation, and generates scaffolds for all 41 active sites in a diverse in-silico benchmark, compared with 16 for previous methods [3]. Parallel efforts have produced designed serine hydrolases [4], metallohydrolases with designed metal sites [5], and enzymes built by catalytic motif scaffolding [6].

The rhetorical work of this subsection is to concede generously. These are strong papers and several share our senior author. The point is not that backbone generation is inadequate — it is that its success relocates the bottleneck downstream, to sequence design and to what happens after the first 96 designs come back weakly active.

Two features of this literature matter for what follows. First, the sequence-design step is treated as a solved subroutine: backbones are passed to ProteinMPNN [7] or LigandMPNN [8] sampled at low temperature, and the resulting sequences are filtered against structure-prediction and geometric criteria. Second, the experimental readout is a small, one-shot test set: RFdiffusion2 reports active candidates after testing fewer than 96 sequences per catalytic mechanism [3]. Both features are appropriate for demonstrating that a backbone generator works. Neither addresses what to do when the resulting enzymes are, as they almost always are, orders of magnitude less active than useful.

2.2 Filtering is not search

The prevailing sequence-design protocol is generate-then-filter. Sequences are sampled from a functionally blind inverse-folding prior, then scored against structure-prediction confidence, backbone recapitulation, catalytic-geometry and preorganization criteria — AlphaFold3 [9], Boltz [10], and ensemble methods such as PLACER, whose introduction substantially improved experimental success rates in serine hydrolase design [4].

This is the paper's central rhetorical move, so state it plainly and early, and make sure the phrasing survives review: a filter is a selection operator, not a search operator.

A filter cannot create a sequence that passes; it can only surface the sequences that a functionally blind prior happened to emit. This has two consequences that the field has largely left implicit. Sampling temperature is lowered to raise the pass rate, which means pass rate is purchased directly with diversity: ProteinMPNN's own report notes that sequence diversity can be increased considerably at higher temperature with only a small decrease in recovery [7], and the corresponding pass-rate cost is why production pipelines sit near $T = 0.1$. And because the survivors are drawn from a narrow neighborhood of the prior's mode, they are strongly correlated with one another — so a library's nominal size overstates the number of genuinely independent attempts it represents.

2.3 Reinforcement learning has reached inverse folding, but not libraries

Aligning generative models to objectives via reinforcement learning is now standard practice, and Group Relative Policy Optimization (GRPO) [11] in particular has proven to be a memory-efficient fit for the regime where rewards are computed by an external oracle rather than learned. This machinery has begun to reach protein sequence design. ProteinZero is an online RL framework for inverse folding that combines structural guidance from ESMFold with a self-derived $\Delta\Delta G$ predictor, adds an embedding-level diversity regularizer to mitigate mode collapse, and reports reducing design failure rates by 36–48% relative to ProteinMPNN, ESM-IF [12], and InstructPLM on the CATH-4.3 benchmark, with success rates above 90% across diverse folds [13]. Notably, its formulation balances three terms — multi-reward optimization, KL-divergence from a reference model, *and* diversity regularization — so the anchor and the diversity mechanism are separate components. Diversity-regularized direct preference optimization has been applied to peptide inverse folding with related motivation [14].

> ProteinZero is the closest prior art to our mechanism and must be confronted in the first paragraph a reviewer reads on this topic, not buried. The differentiation is real but it is about the *objective*, not the optimizer: (i) general folds on CATH vs. enzyme active sites with ligands and catalytic geometry; (ii) sequence-level metrics — designability, recovery, stability — vs. a library-level objective; (iii) a diversity *regularizer* defended as anti-mode-collapse hygiene vs. diversity as a first-class design goal justified by screening economics; (iv) no physical library. If compute allows, running ProteinZero as a baseline arm is the strongest possible version of this argument. **[TODO: decide whether ProteinZero is runnable as a baseline]**

A separate line of work has independently arrived at the structural move our selection experiments test. Explorative Modeling factors the *training loop* rather than the generation procedure: it samples K candidate matches between model generations and data and trains on the best, so that predictions commit to modes rather than blurring them, and reports that scaling exploration acts as a third pretraining axis beyond parameters and data — improving FLOP efficiency 4.1× and sample efficiency 6.2×, and reaching 1.43 FID on ImageNet without guidance [15]. Oversampling candidates and training on a chosen subset is exactly the operation our E18 arms perform, and their observation that the mode-seeking variant "applies no pressure to cover every mode, so on its own it is mode-seeking and can collapse" is an independent derivation of the failure we predict for pure quality-directed selection.

> Cite this as convergent evidence, not as prior art we are extending, and be precise about the disanalogy — a reviewer who knows the paper will check. Their selection criterion is the training objective itself (minimum reconstruction loss against real data), so every candidate can be scored before one is chosen. Ours cannot be: the objective requires a structure prediction, which is the expensive step and the entire reason to oversample. We therefore select on *proxies* — policy log-probability, sequence distance — before any reward is known. That is why a selection rule can lose to random in our setting, which a minimum-loss rule cannot, and it is the reason E18 carries a `random` control at matched oversample: their paper positions exploration as a scaling axis where more candidates inherently help, and reports no ablation separating

selection quality from simply processing more samples.

Read together, these two papers and our own results converge on the same failure from three directions. ProteinZero introduces an embedding-level diversity regularizer specifically to mitigate mode collapse [13]; Explorative Modeling reports that its mode-seeking variant applies no pressure to cover every mode and can collapse on its own [15]; and we find that correcting the RL objective converts a policy that diffuses outward into one that converges onto a narrow solution, with positional entropy falling from ~1.7 to 1.06. Collapse is not incidental to reward-aligned generative training — it is what the corrected objective does by default, and it is the reason a library-level method cannot simply inherit a sequence-level one.

Where we differ is in what the anchor has to do. ProteinZero carries a KL term *and* a separate diversity regularizer; we carry only the KL term, and E13 shows it is sufficient on its own — sweeping its weight moves per-design coverage U_{15} from 510 to 881 monotonically, with total library coverage peaking in an interior plateau. That is a claim about parsimony, not about superiority: their regularizer operates in embedding space and ours does not, and we have not run the two against each other. State it as "one control suffices in our setting", which is defensible, rather than "the regularizer is unnecessary", which we have not tested.

What this line of work does not address is the library. Its objectives and benchmarks are defined per sequence: is *this* sequence designable, stable, recovered. The quantity that determines an experimental outcome is a property of the set — how many distinct, genuinely different candidates clear the bar, and how far apart they are.

2.4 Library design assumes a natural enzyme

A mature literature does optimize libraries, but from the opposite starting point. MODIFY learns from natural protein sequences to infer evolutionarily plausible mutations and predict enzyme fitness, and explicitly co-optimizes predicted fitness and sequence diversity of starting libraries, prioritizing high-fitness variants while ensuring broad sequence coverage [16]. Active Learning-assisted Directed Evolution (ALDE) uses uncertainty quantification over an epistatic five-residue active site and improved the yield of a non-native cyclopropanation product from 12% to 93% in three rounds of wet-lab experimentation [17]. Machine-learning-guided cell-free expression platforms scale the assay side, mapping fitness across large variant sets [18].

MODIFY is the single most important citation for our diversity claim, because it proves the field already accepts that fitness and diversity should be co-optimized in a library. That is helpful — we are not arguing for diversity from scratch. What we are arguing is that every existing method for doing so is unavailable in the de novo regime.

Every one of these methods presupposes what a de novo enzyme lacks. MODIFY's fitness prior is evolutionary, learned from natural homologs; a de novo backbone has no homologs and no alignment. ALDE and active-learning methods require a measurable starting fitness to bootstrap from; de novo designs typically start at or near the detection limit. And both operate over a handful of positions, whereas the mutations that rescue a de novo enzyme are, in our hands, tens of substitutions from the starting design — a combinatorial volume that site-saturation and few-position libraries cannot reach.

2.5 The gap

Nobody has addressed library design for de novo enzymes. The regime is defined by three simultaneous absences: no evolutionary prior, so MSA-based fitness models do not apply; no starting activity, so active-learning loops have nothing to condition on; and a target region of sequence space too far away for local mutagenesis. The field's current answer — low-temperature inverse folding plus hard filters — is not a search procedure at all, and it responds to the difficulty of the regime by narrowing the search precisely when it should be widening it.

Close on the thesis so the reader carries it into the results. Draft formulation, to be tightened once F2 exists: function-aligned online RL converts in-silico filters from a rejection step into a design objective, producing de novo enzyme libraries that simultaneously pass the community's own fidelity criteria at higher rates and span far more sequence space — increasing the number of independent experimental bets per unit of screening effort.

[TODO: gap paragraph — final phrasing of the thesis sentence, once the headline figure is built]

3 Results: function-aligned RL moves the diversity/fidelity frontier

Claim. On an established public benchmark, CLEO produces libraries with the sequence coverage of high-temperature LigandMPNN and the pass rate of low-temperature LigandMPNN — a combination no single temperature achieves.

Why this section leads. The PETase result rests on assays that take months and cannot be re-run by a referee. This one is fully computational, uses a published benchmark and a published deposit, and can be reproduced end to end. It buys the method credibility before any wet-lab claim is weighed.

Status: **scaffold**, and close to **drafted**. Both the CLEO arms and the full temperature ladder are measured. What remains: reconcile the two CLEO rows below (Block 3 quotes the corrected 200-step anchored arm; the notebook's E10 table still carries a superseded pre-fix peak window), rebuild the two PCA panels, fold in the E20 rescue result, and attach the run-to-run error bars from E21.

Block 1 — Setup and the unit of comparison

We evaluate on the atomized motif enforcement (AME) benchmark introduced with RFdiffusion2, comprising 41 catalytic sites scaffolded into 100 backbones each. The published protocol assigns 8 LigandMPNN sequences per backbone and predicts each with 5 Chai models; a prediction succeeds when the motif all-atom RMSD to the design is below 1.5 Å and no backbone atom clashes with the ligand.

The aggregation rule matters more than it looks and is easy to get wrong. We verified it against the deposit rather than assuming: the per-site number reported in `AME_benchmark_summary.csv` is the fraction of the 100 backbones with *at least one* passing prediction out of the 40, matching to zero error across all 41 sites and both methods. It is a best-of-40, not a mean over predictions. Scoring one prediction per sequence and comparing rates therefore understates a comparable method by roughly threefold — in the deposit, only 36 % of the sequences that pass on some Chai model pass on all five. Every number below is on their unit.

- Pilot backbones: three drawn from the `pass` class, at 32/40, 28/40 and 10/40 passing predictions in the deposit.
- Our predictor is AlphaFold3 rather than Chai; best-of-N uses 5 diffusion samples per sequence.
- **Caveat to resolve before submission.** Our implementation of the one predictor-free benchmark quantity, `ligand_dist_des_ncac_min`, does not reproduce the published values: median absolute error 0.42 Å over 40 backbones, *zero* exact matches. Since it shares parsing and superposition code with the motif RMSD, the absolute values are not yet comparable to theirs. Until it is closed, only the backbone that reproduces (M0097) supports a head-to-head claim.
- **This is a metric-definition gap, not a reference-structure gap.** An earlier draft attributed it to our scoring against the idealized backbone where RFdiffusion2 scores against the unidealized, packed design. That explanation is now excluded by direct measurement. We recovered every candidate design structure from the authors' 177 GB benchmark archive — idealized, unidealized, LigandMPNN-packed,

LigandMPNN input backbone — and all four give the same distance to within 0.01 Å while all four miss the published value by ∼0.42 Å. The authors' own `metrics_cache` confirms the mechanism could not have worked: idealization displaces atoms by 0.097 Å mean and 0.274 Å max, an order of magnitude too small. Atom-subset and motif-inclusion variants are likewise excluded.

[TODO: ame — reproduce `ligand_dist_des_ncac_min` exactly; needs `per_sequence_metrics.py`, which is not in the deposit (outputs only). Structure hypothesis falsified 2026-08-15 — do not retry it.]

Scope of this caveat. It bounds only whether our absolute numbers may be set beside the published ones. Every comparison the paper actually makes — CLEO against LigandMPNN, the temperature ladder, the KL sweep, the objective ablation — runs through a single pipeline with a single reference, and none of them is affected.

Block 2 — Low-temperature sampling returns one solution, sampled repeatedly

The incumbent protocol samples LigandMPNN at low temperature. At $n = 96$ on M0097 this yields a high success rate — 58.3 % of sequences pass — but the passing designs are a tight cluster: mean pairwise Hamming distance 40.6 out of $L = 180$, and 162 distinct substitutions among any 15 of them. The library is close to one solution measured many times, and most of the folding budget spent distinguishing its members buys little.

Raising the temperature removes the redundancy and most of the signal with it. At $T = 1.0$ the same backbone reaches mean pairwise distance 126.3 at a success rate of 4.2 %. Diversity alone is not the objective; a library that cannot be filtered is not a library.

Two numbers here were wrong in earlier drafts and are corrected. $T = 1.0$ was reported as *zero* passing designs; that came from $n = 8$, and at $n = 96$ it passes 4.2 % of the time. And the low-temperature arm was described as collapsing to a single cluster "at every threshold down to 50 % identity" — true at 70 %, but at 90 % every passing design is its own cluster and the statistic simply counts designs. Cluster counts are threshold-dependent and are no longer quoted; mean pairwise Hamming and the distinct-substitution count U are threshold-free and replace them throughout.

Block 3 — cleo occupies the corner neither temperature reaches

Sampling temperature traces a frontier: success falls as diversity rises, and every point on it is available to the incumbent by turning one knob. The claim of this section is that CLEO sits above that frontier, and the measurement is in Table 1.

Interpolating the baseline frontier at the anchored arm's diversity, the incumbent succeeds 14 % of the time where CLEO succeeds 56.7 % — **3.98**×. Read the other way, at CLEO's success rate the incumbent reaches mean pairwise distance 59 against CLEO's 114.4 — **1.93**×. The anchored arm also carries more distinct substitutions per passing design than any baseline ($U_{15} = 808$ against 608 at T=0.7), so this is a genuine exploration advantage and not only a higher yield.

The unanchored arm is the instructive contrast. It reaches the highest success rate in the table, 83.1 %, and the *worst* per-design coverage of any CLEO arm — $U_{15} = 510$, below T=0.7. Left unconstrained the policy converges onto a narrow solution. The KL weight against the base model is what holds the distribution open, and it is the control the library application needs.

Sweeping that weight over $\{0, 0.005, 0.01, 0.02, 0.05, 0.1\}$ shows it is a well-behaved control with an interior optimum rather than a knob that has to be tuned. Success falls monotonically, 83 % to 10 %, and per-design coverage rises monotonically, U_{15} 510 to 881; the total distinct substitutions a library actually contains is the product of the two and is flat within noise across $w \in [0.01, 0.05]$ (2480 to 2501) before collapsing to 1743 at $w = 0.1$, where too few designs pass for per-design

Table 1: M0097, per-sequence success rate on best-of-5 AF3 predictions, with diversity measured over the *passing* designs only — a library contains only what survives filtering. U_{15} is the number of distinct (position, residue) substitutions in a library of 15 passing designs, rarefied over 300 draws so arms with different yields are comparable. Baselines are $n = 96$; CLEO arms are the final 40 steps of a 200-step run.

Arm	**Success**	**Mean pairwise Hamming**	U_{15}	**Median RMSD (Å)**
LigandMPNN T=0.1	58.3 %	40.6	162	1.42
LigandMPNN T=0.2	66.7 %	47.8	205	1.37
LigandMPNN T=0.3	58.3 %	57.9	274	1.31
LigandMPNN T=0.5	34.4 %	80.5	438	1.70
LigandMPNN T=0.7	26.0 %	100.5	608	1.88
LigandMPNN T=1.0	4.2 %	126.3	—	5.59
CLEO, no anchor	83.1 %	76.6	510	1.07
cleo, KL anchor	56.7 %	**114.4**	**808**	1.44

exploration to accumulate. The value used throughout, $w = 0.02$, sits in the middle of that plateau, so the reported numbers do not depend on having found a sharp optimum.

Figure 1 shows why the success rate alone is not enough: on M0097 every CLEO arm has its *median* below the 1.5 Å cutoff, so the mode has moved rather than the tail merely reaching across — which is precisely what fails to hold on the near-miss backbones of Block 6.

Block 4 — The training trajectory is the library

This is the reframing that removes a hyperparameter. Rather than adding an explicit diversity term to the reward, we treat every rollout sampled during training as a library candidate. Those sequences were already folded and scored to compute the reward, so the candidate pool is free: the library is a byproduct of training, not an additional expense.

Across 200 GRPO steps at 16 sequences per step, M0097 accumulates 3200 scored designs. Under the anchored policy 1349 of them are distinct passing sequences carrying 3045 distinct substitutions — 89 % of the $19L = 3420$ ceiling.

	cleo, no anchor		**cleo, KL anchor**	
Cumulative folds	**Unique passing**	**Substitutions**	**Unique passing**	**Substitutions**
400	75	1483	47	1382
800	284	2294	160	2045
1600	893	2805	510	2757
2400	1560	2974	903	2958
3200	2229	3040	1349	3045

The distinct-substitution count is genuinely saturating here, at 89 % to 89 % of the ceiling, while the count of distinct passing sequences is still climbing steeply. The library is not running out of designs; it is running out of *single* substitutions to explore, which is the expected behaviour once most positions have been visited and further diversity must come from combinations.

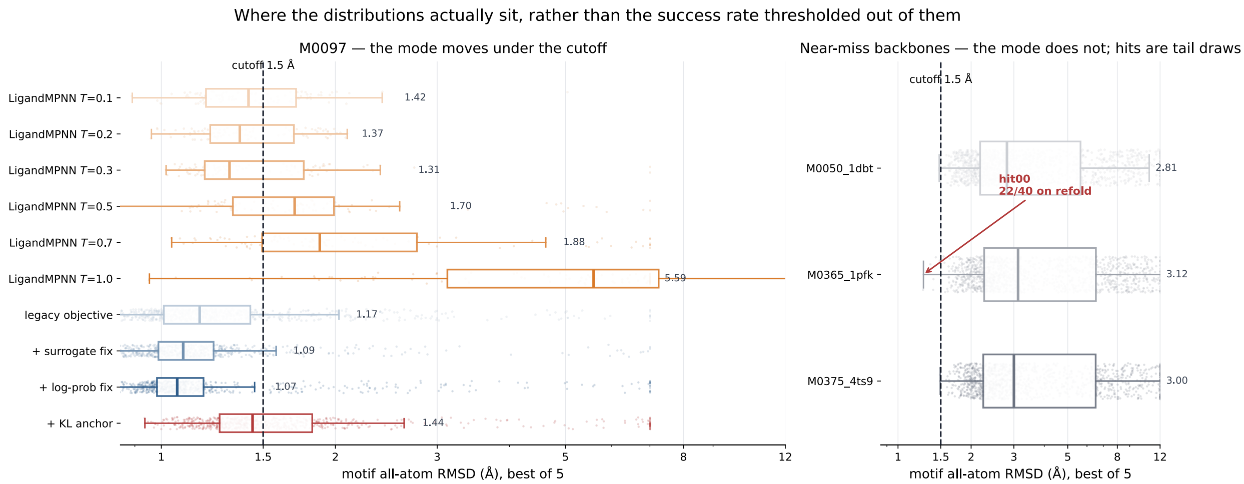

Figure 1: **Where the distributions actually sit, rather than the success rate thresholded out of them.** Each row is one arm's per-design best-of-5 motif RMSD: box for median and quartiles, jittered strip behind it for the shape the box hides. Log axis, because a linear one lets the single worst arm compress everything that matters into the left fifth. **Left, M0097:** every CLEO arm has its *median* below the cutoff (1.07 Å to 1.44 Å) — the mode has moved, not just the tail — against 1.31 Å to 1.42 Å for low-temperature LigandMPNN and 5.59 Å at T=1.0. The non-monotonicity noted in E10 is visible directly: T=0.3 has a *lower* median than T=0.1. **Right, the three near-miss backbones:** medians of 2.81 Å to 3.12 Å with the cutoff far out in the left tail. The one design that survived 40 refolds sits at the extreme edge of its own distribution. Encoding: both ladders are ordinal, so each is a single hue ramped light to dark rather than arbitrary categorical colours; the anchored arm keeps an accent hue as the highlighted result.

This supersedes an earlier version of this table reporting 97 passing designs from 3200 folds, a factor of 14 to 23 lower. That run used the pre-audit objective, and its curve flattened because the policy decayed rather than because sequence space was exhausted — the earlier draft said so and predicted rank normalisation would keep it climbing, which is what happened. The pool numbers here are best-of-5, matching Block 3; the superseded ones were single-sample and were not comparable to it.

Block 4b — Where in sequence space the passing designs sit

Generated by `paper/figures/ame_pca.py`, which writes two figures because one projection cannot honestly carry both claims. The shared projection is fit across arms within a backbone — a per-arm PCA places every cloud at the origin and destroys the comparison — but deliberately not across backbones, where a global fit puts 99 % of PC1 variance on backbone identity.

These two panels are stale and must be regenerated. They were built from the pre-audit runs and from baseline arms holding 8 sequences each. Both inputs have been replaced: the objective is fixed and the baselines are now $n = 96$. The drift control below remains the right control and its conclusion is unlikely to change, but no number in these captions should be quoted until they are rebuilt.

Explained variance is low (PC1 5 % to 9 %), as expected for one-hot protein sequence, so the projection is a visual aid; every quantitative diversity claim in this section uses Hamming distance and substitution counts directly.

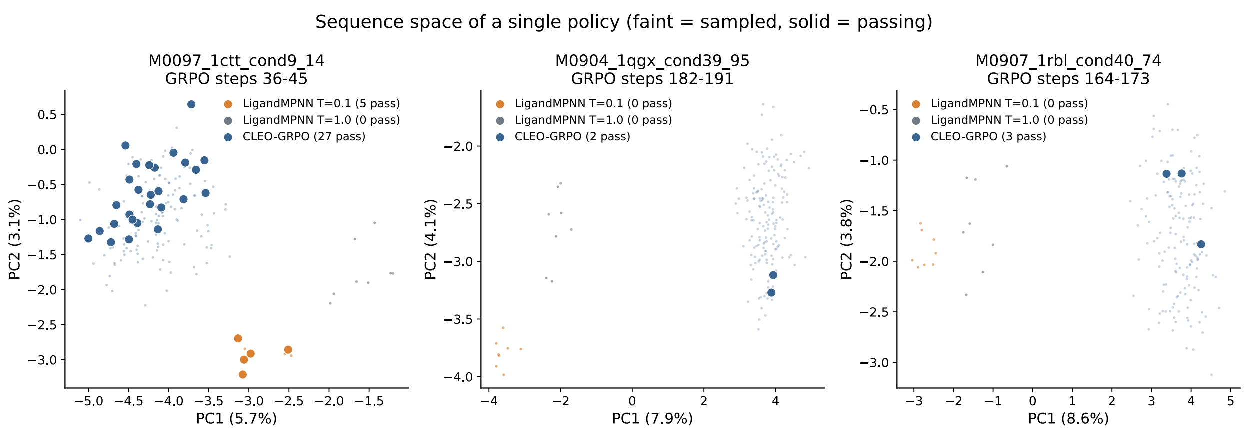

Figure 2: **Sequence space occupied by each design method on the AME pilot backbones.** One PCA per backbone, fit on every sequence that backbone contributes across all three arms, so the arms share one projection and the clouds are directly comparable. Faint points are sampled designs, solid points are designs that pass the benchmark criterion (motif all-atom RMSD < 1.5 Å, no ligand clash). CLEO is restricted to its peak-performing 10-step window so that three *policies* are compared at one point in training. On M0097 the low-temperature arm (orange) forms a tight knot displaced far along PC1 — five passing designs that are a single cluster at every identity threshold down to 50 % — while CLEO's passing designs are spread throughout a broad cloud. T=1.0 (grey) reaches comparable spread with zero passing designs. Projections are deliberately *not* shared across backbones: a global fit places 99 % of PC1 variance on backbone identity, collapsing each backbone to its own blob and hiding the arm differences entirely. Explained variance is low (PC1 5 % to 9 %), as expected for one-hot encoded protein sequence; the projection is a visual aid and every quantitative claim in the text uses Hamming distance and cluster counts directly. Baseline arms hold 8 sequences each, so part of their compactness is small-n.

[TODO: ame — rebuild both PCA panels on the corrected runs at matched n]

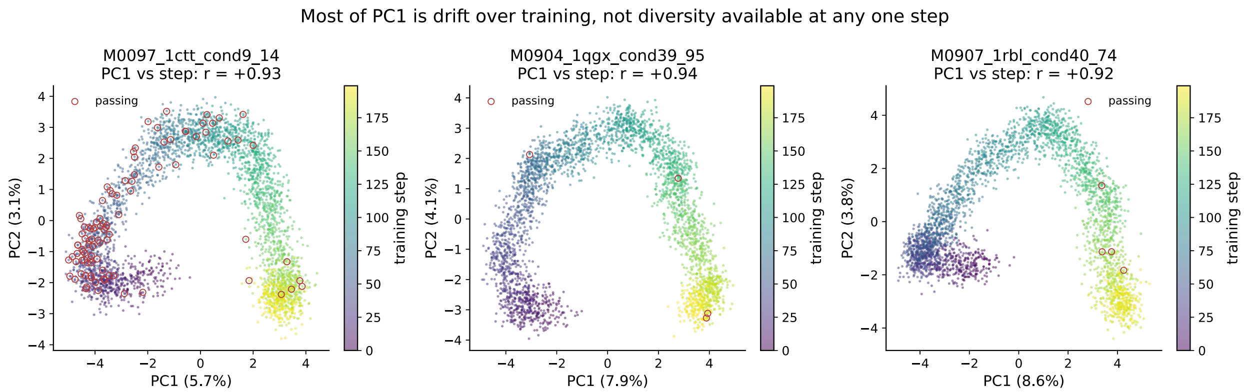

Figure 3: **Most of PC1 is drift over training, not diversity available at any one step.** The full GRPO trajectory for each backbone, in the same projection as Figure 2, coloured by training step; passing designs are circled. PC1 correlates with training step at $r = +0.92$ to $+0.94$ on all three backbones. This distinction is load-bearing rather than cosmetic: spread accumulated by drifting across training is genuine library diversity when the library is assembled from the whole run, which is how ours is assembled, but it is *not* evidence that any single policy is diverse. Presenting the pooled projection alone would conflate the two and overstate the result, so the axis is shown rather than hidden.

Block 5 — The temperature frontier

Measured, and reported in Table 1. Two predictions stated in advance were tested.

The dangerous intermediate temperatures do not close the gap. $T = 0.2$ and $T = 0.3$ were named as the points that would refute a frontier shift if either reached CLEO's coverage at CLEO's success rate. Neither does: both sit below 0.6 of the anchored arm's diversity. The frontier shift stands.

One prediction was wrong. The trade-off was expected to be monotone in temperature; it is not. $T = 0.2$ is the *most* successful arm in the table at 66.7 %, above $T = 0.1$'s 58.3 %, while also being more diverse. $T = 0.2$, not $T = 0.1$, is therefore the incumbent to quote at the low-diversity end, and any comparison against $T = 0.1$ alone flatters us.

> Still outstanding from this block's original plan: the saturation fold count for low-temperature LigandMPNN — the point past which it re-emits sequences it has effectively already produced. The CLEO side of that comparison is Block 4; the baseline side needs a cumulative-distinct curve at matched budget, which the $n = 96$ sweep can supply but does not yet.

Block 6 — Rescue of backbones the benchmark cannot solve

Three backbones sit at 1.61 Å to 1.69 Å best-of-40 in the deposit — zero passing predictions, but only ∼0.15 Å from threshold. A pass on any of them is a design the published pipeline cannot produce, with no reproduction confound, because there is no positive result of theirs for us to fail to reproduce.

All three were trained for 75 steps at batch 16. Nine designs cleared the criterion. **Each was then refolded 40 times — the published per-backbone budget — and one survives.** M0365 `hit00` passes 22 of 40 predictions with a *median* RMSD of 1.470 Å, below the cutoff rather than touching it, on a backbone the deposit scores 0/40 with a best of 1.69 Å. That is a design the published pipeline does not produce.

The other eight do not survive. Five pass zero of 40; three pass one or two. Training-time margin predicts this almost perfectly: the single design with a margin above 0.03 Å inside the cutoff is the single design that replicates, and everything at 0.003 Å to 0.023 Å was a tail draw from a distribution whose median sits near 3 Å (Figure 1, right).

The honest claim is one rescue, not nine, and the general lesson is worth more than the rescue: a best-of-N filter applied to a distribution whose mode is far from the cutoff selects excursions, not solutions. A design that truly passes 1 in 40 times is flagged by a best-of-5 filter about 12 % of the time it is folded. Every headline number that leaves this section should be replicated the way `hit00` was, and in-training pass rates should be described as counts of best-of-N events rather than of reproducible designs.

Two caveats remain open. The fold budget is not matched — 1200 designs at best-of-5 against the deposit's 8 sequences at 5 models — so a compute-matched comparison is still owed. And the idealised-versus-unidealised reference structure question from Block 1 applies here as it does everywhere.

4 Results: fragment recombination builds the library

Claim. Designs from the trained policy can be cut at fixed boundaries and recombined across parents, so that $\sum_i |F_i|$ synthesized DNA parts yield $\prod_i |F_i|$ testable variants — and the recombinants keep enough of the parental pass rate that the resulting library is worth building.

Status: needs data. This is the one section of the paper with no data behind it yet. Every other claim in the document is either measured or a rewrite of something measured. Written out in full here so the experiments it needs are unambiguous.

Why it is the load-bearing section. Section 3 buys a better *distribution*; on its own that is a sampling result. This section is what converts a distribution into something you can physically order, and it is the only reason the PETase libraries in Section 5 can be the size they are. Without it the paper is a benchmark result with an application attached.

Block 1 — The combinatorial argument, stated precisely

- Set up the notation: L residues split into k fragments at fixed boundaries $[s_i, e_i]$; F_i the set of distinct sequences observed at fragment i across the policy's output; a recombinant is one choice per slot, joined.
- The arithmetic: $\sum_i |F_i|$ parts, $\prod_i |F_i|$ recombinants. With $k = 5$ and $|F_i| = 40$ that is 200 parts and 1.0×10^8 variants. **State the actual numbers used, not this example.**
- Golden Gate assembly is what makes the product physically reachable: one pot, orthogonal overhangs at the boundaries, no per-variant synthesis. Methods carries the adapter design.
- **The assumption the whole section rests on, named up front:** fragments are treated as if they contribute independently to foldability. They do not, exactly — the whole point of an inverse folding model is that residues are coupled. Block 3 measures how badly the approximation fails, and that measurement is the result.

[TODO: fragments — state the actual k, $|F_i|$ and library size used]

Block 2 — Where the boundaries go

- Boundaries are chosen on the structure, not the sequence: cut in loops and between secondary-structure elements, never through the active site.
- **Pre-register the comparison:** boundaries placed on secondary-structure breaks should retain more pass rate than boundaries placed at evenly spaced sequence positions. If they do not, the structural rationale is decorative and should be dropped from the paper rather than defended.
- Report the fixed-residue set — catalytic residues held constant across all fragments, so every recombinant retains the active site by construction.

[TODO: fragments — boundary placement rule + the structural-vs-uniform control]

Block 3 — Do recombinants fold? (the experiment that must be run)

This is the measurement the section exists for. Everything above is setup and everything below is consequence. Design, in the order it should be run:

1. Sample n designs from a trained CLEO policy with no selection applied, on an AME backbone where the parental pass rate is already known and high (M0097 is the obvious choice — it is the backbone every other measurement in the paper uses).
2. Split each at the fragment boundaries. Keep only fragments drawn from *passing* parents, since that is what one would actually order.
3. Draw recombinants uniformly from the product space, excluding any recombinant that reproduces a parent exactly.
4. Fold best-of-5 and score on the identical code path as Section 3. **Same pipeline, same reference, or the comparison to the parental rate is not a comparison.**

The endpoint is retention: recombinant pass rate divided by parental pass rate. Everything else is secondary.

Pre-registered predictions, written before the run:

- Retention is well above the rate for recombining across *random* MPNN samples at matched diversity. If it is not, fragmentation is adding nothing that temperature would not.
- Retention falls monotonically with the number of slots swapped away from a single parent. A recombinant differing at one slot should behave nearly like its parent; one differing at all k should be worst. A flat curve would mean the fragments are effectively independent, which would be a stronger result than we expect and should be reported as such.
- Recombinants reach mutational combinations absent from the parental set — measured as U over the recombinant library against U over the parents at matched design count.

What kills the claim: retention below the point where the library is filterable in practice. If the parental rate is 57 % and recombinants pass at 2 %, the product space is nominal rather than real, and the honest paper reports the fragment library as a screening funnel with a stated hit rate instead of as a 10^8-member library. Decide the threshold now, not after seeing the number.

[TODO: fragments — RUN THIS. Selection-free sample → split → recombine → fold best-of-5 → retention curve vs. slots swapped]

Block 4 — Relation to manufacturing-aware generative models

(author?) [1] is the nearest prior art and the section must engage it directly rather than in passing. Their move is to make the generative model manufacturing-aware from the outset — factorized to match what combinatorial DNA chemistry can physically sample — and they reach $\sim 10^{16}$ designs for antibodies, T-cell antigens and Taq polymerase.

The honest framing. They get there first and at larger scale. What is different is the regime, and it is a real difference rather than a hedge:

- **Their objective comes from a natural distribution.** 300 million observed human antibodies define what a good sample looks like, so the model can be constrained to the chemistry from the start without giving anything up. For a de novo enzyme there is no such distribution — the backbone has no natural homologs — so the objective has to be constructed, which is what Section 3 does.
- **Order of operations.** They factorize first and train the factorized model; we train an unfactorized policy for catalytic geometry and factorize afterwards. That makes retention (Block 3) our problem and not theirs — their samples are manufacturable by construction, ours have to be shown to survive the cut.

- **Scale is not the axis we win on and the section should say so plainly.** 10^{16} is out of reach here and it does not need to be reached: a 10^{8} library that is enriched for catalytic geometry is a different object from a 10^{16} library that is not.

Do not write this block as a competitive comparison. The two results compose — a manufacturing-aware architecture trained against a catalytic reward is the obvious next thing, and saying so is more useful than claiming a win.

[TODO: fragments — write Block 4 once Block 3 lands; needs the library-size and retention numbers to be concrete about the 10^{8} vs 10^{16} point]

5 Results: three rounds of closed-loop optimization on a de novo PETase

Claim. Three successive libraries, each designed using the measured activity of the previous one, take a de novo PETase from a weak starting design to **[250]**× faster — above every characterized natural enzyme on the same probe.

Status: needs data. The assays exist; the numbers are not yet in this repository. Each block below names exactly what file or measurement it needs, so the section can be filled in without re-deriving the structure.

Why three rounds and not one big library. The escalation is the argument. Round 1 uses an easy fluorogenic probe because a cold-start library needs a permissive assay to produce any signal at all. Round 2 moves to a substrate closer to real PET *and* gets larger, which is only affordable because of Section 4. Round 3 is the closed loop proper: the design objective now contains a surrogate trained on rounds 1 and 2. Each round changes exactly one thing, which is what makes the final number attributable.

Block 1 — Starting point and the assay ladder

- The de novo PETase backbone: origin, catalytic triad, why it is a genuine de novo design and not a remodelled natural PETase.
- Baseline activity of the starting design — the denominator for every fold-improvement number in this section. **State it once, here, with units, and refer back to it.**
- The three probes in order of stringency: the easy fluorogenic probe, the PET-proximal probe, and bulk PET. Say up front which claims are on which probe — the single most likely way this section misleads a referee is by quoting a fold-improvement from one probe next to a comparison from another.
- Comparator panel: which natural and engineered PETases were assayed *in-house on the same probe*. Literature values are not a comparison and should not be presented as one.

[TODO: petase — backbone provenance, starting activity with units, probe ladder, comparator panel]

Block 2 — Round 1: pilot library on the easy probe

- Library 1: designed by function-aligned RL sampling, no surrogate — the reward is computational only.
- Size, fragment structure, mutation-distance distribution from the starting design.
- Activity distribution: hit rate, best variant, fold over start.
- **What round 1 has to establish:** that the library contains signal at all, and that the signal spans enough range to train on. A library where everything is dead teaches the surrogate nothing.

[TODO: petase — round 1 library composition and activity distribution]

Block 3 — Round 2: larger library, harder probe

- What changed from round 1, and what deliberately did not.
- Library 2 size — this is where the fragment combinatorics pay off, so the number should be stated against the count of DNA parts actually ordered.
- Activity on the PET-proximal probe: hit rate, distribution, best variant.
- **Expect the hit rate to fall** moving to the harder probe. Report it falling. A round that got harder and easier at the same time invites the reader to wonder what else changed.

[TODO: petase — round 2 library and activity on the PET-proximal probe]

Block 4 — Round 3: the surrogate closes the loop

- Surrogate: MLP ensemble on (sequence, activity) pairs pooled from rounds 1 and 2. Architecture, training split, held-out calibration.
- **Report held-out predictive performance before reporting what it designed.** A surrogate that cannot rank held-out variants did not cause the improvement, whatever the improvement turns out to be.
- Surrogate added as a reward step in the RL pipeline; library 3 composition and mutation distance.
- Activity distribution of library 3 against library 2 on the same probe — **the whole-distribution comparison, not just the best variant.** A shifted distribution is evidence the loop closed; a single outlier is evidence of a bigger library.

[TODO: petase — surrogate calibration, library 3, distribution shift vs. round 2]

Block 5 — The best construct

- Full kinetics on the top variant: k_{cat}, K_{M}, $k_{\text{cat}}/K_{\text{M}}$, against the starting design under identical conditions.
- The [**250**]× figure: state precisely which quantity, which probe, which comparator.
- Head-to-head against the natural and engineered comparators on that probe.
- Mutation distance from the starting design, and how far outside SSM range that is — this closes the loop back to Section 1's framing.
- Structural rationalization if a structure exists; clearly marked as post-hoc if so.

[TODO: petase — top-variant kinetics table + comparator head-to-head]

Block 6 — What this does not show

- Bulk PET. If activity on solid substrate does not follow the probe results, say so in the main text rather than the supplement, and attribute it to substrate accessibility only if there is evidence for that attribution.
- Cold start. The loop needs a first library with measurable spread; this section does not show how to get one on a system where round 1 comes back flat.
- One backbone, one chemistry. The generality claim rests on Section 3, which is 41 sites, not on this section.

[TODO: petase — limitations, including the bulk-PET gap]

6 Discussion

Science research articles end with a short discussion / outlook rather than a Conclusion section. Keep this tight (∼250–400 words). Write last, after the abstract. The four paragraphs below mirror the three-move spine plus limits.

P-A — The central takeaway

- The active-variant region for a de novo enzyme is farther from the starting design than SSM can reach and more specific than temperature sampling can hit. Both halves of that sentence are measured in this paper, and neither is measured by prior work.
- The resolution is not a better sampler or a bigger library on its own. It is a sampler trained against function, factorized into parts that can be physically assembled, and closed against measured activity. Each piece is necessary; state why.

[TODO: discussion p1 — central takeaway]

P-B — What each move established

- **Frontier.** A KL-anchored RL policy occupies a region of the diversity/fidelity plane that no temperature reaches. The anchor is the mechanism, not a detail — unanchored, the policy collapses onto a narrow solution with worse per-design coverage than the baseline.
- **Fragments.** Recombination converts that distribution into a library exponentially larger than the parts ordered, at a measured cost in pass rate.
- **PETase.** Three rounds, each designed on the last round's data, ending above characterized natural enzymes on the same probe.

[TODO: discussion p2 — what each move proves]

P-C — Honest limitations

- **Bulk PET.** Probe activity is not plastic degradation. Do not let this paragraph soften.
- **Run-to-run variance.** Single-run comparisons of pass rate on this benchmark are not interpretable at the effect sizes usually of interest; two draws of one config moved 21 points. Every claim retained in this paper either clears that width or is replicated. *Cite the notebook's E17/E19/E21 record for the measurement.*
- **The reward and the evaluation share a predictor.** Motif RMSD under AF3 appears on both sides of the frontier result. A held-out oracle remains the outstanding control, and the paper should say so rather than wait to be asked.
- **Selection during training does not help.** Five experiments looked for a rule that beats random selection of rollouts and none found one. Worth one sentence as a negative result others need not repeat.
- **Cold start.** The loop needs a first library with measurable spread.

[TODO: discussion p3 — limitations]

P-D — Outlook

- The obvious composition: a manufacturing-aware architecture [1] trained against a catalytic reward, rather than factorizing after the fact as we do here. That gets the scale of the former and the objective of the latter, and it is the next experiment rather than a distant prospect.
- More rounds, and rewards that mix computational and measured signal continuously rather than once per library.
- Other chemistries and substrate classes, including bulk polymers.

[TODO: discussion p4 — outlook]

7 Methods: shared pipeline

Shared methodology underlying all three applications. Per-application specifics live in the following subsections. Do not write this out in detail before the Results numbers exist — Methods text written against placeholder numbers gets rewritten.

Block 1 — RL fine-tuning of ProteinMPNN with GRPO

- Reference `cleo.design.train_policy` and `cleo.design.utils.grpo`.
- Reward pipeline composition (`UniversalReward`, ordered metric steps, normalized weighted aggregation).
- Cite GRPO [11] and ProteinMPNN [7].

[TODO: methods overview — GRPO + reward pipeline]

Block 2 — Fragment-based combinatorial libraries

- Sampling, fragment splitting, combinatorial recombination.
- DNA assembly via Golden Gate adapters.

[TODO: methods overview — fragment library construction]

Block 3 — Surrogate models for closed-loop optimization

- MLP ensemble with Gaussian NLL.
- Acquisition function (BatchUCB + diversity).
- Used as a reward step inside the RL pipeline.

[TODO: methods overview — surrogate ensemble + acquisition]

Block 4 — Batch-consensus mutation-diversity reward

- Per-position modal AA across the batch (ties multi-modal).
- Two metrics: `consensus_divergence` (count) and `fractional_divergence` ($1/k$ weighted, penalizes secondary-mode collapse).
- Reference `cleo.design.utils.mutation_diversity`.
- Contrast with the embedding-level diversity regularizer of ProteinZero [13] and diversity-regularized DPO [14]: ours acts in mutation space against a named reference sequence, theirs in embedding space.

[TODO: methods overview — diversity reward definition]

7.1 Methods: PETase

- **Block 1 — Library 1 design configuration.** Backbone, fixed catalytic residues, library 1 reward pipeline (Boltz oracle [10] + PETase metrics + Hamming distance to reference).
- **Block 2 — Surrogate training.** Featurization, train/val split, ensemble size, calibration check.
- **Block 3 — Library 2 with surrogate-as-reward.** Reward pipeline composition for library 2.
- **Block 4 — Activity assays.** PET-trimer probe, PET-dimer, bulk PET attempt.

[TODO: petase methods — library 1 design configuration] [TODO: petase methods — surrogate] [TODO: petase methods — library 2 design configuration] [TODO: petase methods — assays]

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